GATE ME 90 Days Challenge

Day 83/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Full Syllabus	Full-Length Mock 5	15 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on a mixed full-syllabus warm-up for the fifth full-length mock: instrument analogies, train-and-platform speed, inverse Laplace transforms, the trapezoidal rule, lifting-machine efficiency, shear in I-beams, shaft torsion, the steering-gear condition, natural frequency from static deflection, Petroff's bearing equation, hydraulic turbine power, composite walls, the Carnot refrigerator, non-consumable electrodes and pull production control. Try them in **60 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 11 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 65 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

Thermometer : Temperature :: Anemometer : _____

(A) Pressure
(B) Humidity
(C) Wind speed
(D) Rainfall

Q2 NAT

A train 180 m long crosses a platform 120 m long in 15 seconds. The speed of the train (in km/h) is _____

Q3 MCQ

The inverse Laplace transform of $F(s) = 1/[s(s + 1)]$ is

(A) e^{-t}
(B) $1 - e^{-t}$
(C) $1 + e^{-t}$
(D) $t\,e^{-t}$

Q4 NAT

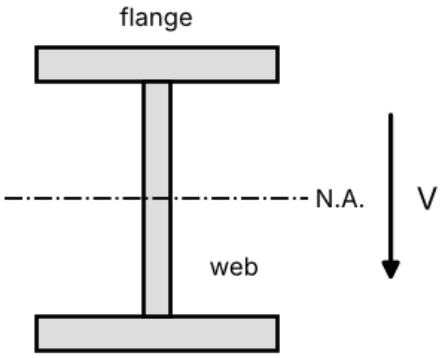
The integral of x^3 from $x = 0$ to $x = 1$ is evaluated by the trapezoidal rule with four equal strips ($h = 0.25$). The value obtained (round off to three decimal places) is _____

Q5 NAT

In a lifting machine with a velocity ratio of 20, an effort of 50 N just lifts a load of 800 N. The efficiency of the machine (in %) is _____

Q6 MCQ

The I-section beam shown carries a vertical shear force. Which statement about the transverse shear stress in the section is correct?



(A) It is greatest at the top and bottom fibres

(B) It is uniform over the whole depth

(C) The flanges carry most of the shear force

(D) The web carries most of the shear force, and the stress is greatest at the neutral axis

Q7 NAT

A solid circular shaft of 40 mm diameter is subjected to a torque of 1 kN·m. The maximum shear stress in the shaft (in MPa, round off to one decimal place) is _____

Q8 MCQ

For correct steering (pure rolling of all four wheels about a common centre) of a car with wheelbase b and distance c between the front stub-axle pivots, the angles turned by the outer front wheel (φ) and the inner front wheel (θ) must satisfy

(A) $\cot\varphi - \cot\theta = c/b$

(B) $\tan\varphi - \tan\theta = c/b$

(C) $\cot\varphi + \cot\theta = c/b$

(D) $\cos\varphi - \cos\theta = b/c$

Q9 NAT

A machine mounted on springs deflects them statically by 2.5 mm under its own weight. Taking $g = 9.81\ m/s^2$, the natural frequency of its vertical vibration (in Hz, round off to two decimal places) is _____

Q10 MCQ

Petroff's equation for the friction coefficient of a journal bearing applies to

(A) a heavily loaded bearing with a large eccentricity

(B) a bearing working with boundary lubrication

(C) a lightly loaded bearing in which the journal is taken as concentric with the bearing

(D) a rolling-contact (ball) bearing

Q11 NAT

A hydraulic turbine works under a net head of 100 m with a flow of 2 m^3/s, and its overall efficiency is 90 %. Taking $g = 9.81\ m/s^2$, its shaft power output (in MW, round off to two decimal places) is _____

Q12 MCQ

A composite wall has two layers of equal thickness in perfect contact, and the conductivity of layer 1 is twice that of layer 2. In steady one-dimensional conduction the total temperature drop across the wall is 90 °C. The temperature drop across layer 2 is

(A) 30 °C
(B) 60 °C
(C) 45 °C
(D) 90 °C

Q13 NAT

A Carnot refrigerator keeps a cold space at −23 °C while rejecting heat to surroundings at 27 °C. Its coefficient of performance is _____

Q14 MCQ

Which one of the following arc welding processes uses a non-consumable electrode?

(A) Shielded metal arc welding (SMAW)
(B) Submerged arc welding (SAW)
(C) Gas metal arc welding (GMAW/MIG)
(D) Gas tungsten arc welding (GTAW/TIG)

Q15 MCQ

Which one of the following is a 'pull' system of production control?

(A) Kanban (just-in-time)
(B) Material requirements planning (MRP)
(C) Scheduling production in advance from a sales forecast
(D) Releasing fixed economic batch quantities on a set calendar

Answer Key

Q	Type	Answer
Q1	MCQ	(C) Wind speed
Q2	NAT	72
Q3	MCQ	(B) $1 - e^{-t}$
Q4	NAT	0.266 (accept 0.265 to 0.267)
Q5	NAT	80
Q6	MCQ	(D) The web carries most of the shear force, and the stress is greatest at the neutral axis
Q7	NAT	79.6 (accept 79.2 to 80.0)
Q8	MCQ	(A) $\cot \varphi - \cot \theta = c/b$
Q9	NAT	9.97 (accept 9.92 to 10.02)
Q10	MCQ	(C) a lightly loaded bearing in which the journal is taken as concentric with the bearing
Q11	NAT	1.77 (accept 1.75 to 1.78)
Q12	MCQ	(B) 60 °C
Q13	NAT	5
Q14	MCQ	(D) Gas tungsten arc welding (GTAW/TIG)
Q15	MCQ	(A) Kanban (just-in-time)

Detailed Solutions

Q1 Answer: (C) Wind speed

Concept: The pair links an instrument to the quantity it measures.

A thermometer measures temperature. A barometer measures pressure, a hygrometer humidity and a rain gauge rainfall.

An anemometer measures **wind speed**, option (C).

Exam tip: State the relation in a short sentence ('X measures Y') and test each option in the same sentence.

Q2 Answer: 72

Concept: To cross a platform the train must cover its own length plus the platform length.

Distance = 180 + 120 = 300 m; speed = 300/15 = 20 m/s.

20 m/s × 18/5 = **72 km/h**.

Exam tip: Crossing a pole or a person needs only the train's own length; crossing a platform or bridge needs both lengths.

Q3 **Answer: (B) $1 - e^{-t}$**

Concept: Split into partial fractions, then use $L^{-1}\{1/s\} = 1$ and $L^{-1}\{1/(s + a)\} = e^{-at}$.

$1/[s(s + 1)] = 1/s - 1/(s + 1)$.

So $f(t) = \mathbf{1 - e^{-t}}$, option (B).

Exam tip: Check with the initial value theorem: $sF(s) \to 0$ as $s \to \infty$, so $f(0) = 0$, which only $1 - e^{-t}$ satisfies.

Q4 **Answer: 0.266 (accept 0.265 to 0.267)**

Concept: Trapezoidal rule: $I \approx (h/2)[y_0 + 2(y_1 + y_2 + y_3) + y_4]$.

y at x = 0, 0.25, 0.5, 0.75, 1: 0, 0.015625, 0.125, 0.421875, 1.

$I \approx 0.125 \times [0 + 2(0.5625) + 1] = 0.125 \times 2.125 = \mathbf{0.266}$ (exact value 0.25).

Exam tip: For a function that curves upward (convex), the trapezoidal rule always over-estimates the integral.

Q5 **Answer: 80**

Concept: Efficiency = mechanical advantage ÷ velocity ratio, where MA = load/effort.

MA = 800/50 = 16.

$\eta = 16/20 = 0.80 = \mathbf{80\ \%}$.

Exam tip: An ideal (frictionless) machine would lift VR × effort = 1000 N; the 'lost' 200 N of load is the effect of friction.

Q6 **Answer: (D) The web carries most of the shear force, and the stress is greatest at the neutral axis**

Concept: $\tau = VQ/(Ib)$: the shear stress depends on the first moment Q of the area beyond the level and on the width b at that level.

Q is zero at the top and bottom fibres and greatest at the neutral axis, so τ is greatest there.

Where the width drops from the flange width to the thin web, τ jumps up sharply; the web therefore carries most (typically 90–95 %) of the vertical shear: option **(D)**.

Exam tip: For design, the web shear stress of an I-beam is often taken as V/(web depth × web thickness).

Q7 **Answer: 79.6 (accept 79.2 to 80.0)**

Concept: For a solid circular shaft, $\tau_{max} = 16T/(\pi d^3)$, at the outer surface.

$T = 1 \times 10^6$ N·mm; $d^3 = 64\,000$ mm³.

$\tau_{max} = 16 \times 10^6/(\pi \times 64\,000) = \mathbf{79.6\ MPa}$.

Exam tip: Keep N and mm together so the answer comes out directly in MPa (N/mm²).

Q8 **Answer: (A) $\cot\varphi - \cot\theta = c/b$**

Concept: All wheel axes must meet at one point (the instantaneous centre) on the line of the rear axle.

If that point is at distance x from the inner pivot along the rear axle line: $\cot\theta = x/b$ and $\cot\varphi = (x + c)/b$.

Subtracting: **$\cot\varphi - \cot\theta = c/b$**, option (A). The outer wheel turns through the smaller angle.

Exam tip: The Davis gear satisfies this condition exactly at all angles; the Ackermann linkage satisfies it only at three positions, but is simpler (only turning pairs).

Q9 **Answer: 9.97 (accept 9.92 to 10.02)**

Concept: $mg = k\delta$ gives $k/m = g/\delta$, so $f_n = (1/2\pi)\sqrt{(g/\delta)}$.

$g/\delta = 9.81/0.0025 = 3924\ s^{-2}$; $\sqrt{3924} = 62.64$ rad/s.

$f_n = 62.64/(2\pi) =$ **9.97 Hz**.

Exam tip: A handy shortcut is $f_n \approx 15.76/\sqrt{\delta}$ with δ in mm (here $15.76/1.581 = 9.97$ Hz).

Q10 **Answer: (C) a lightly loaded bearing in which the journal is taken as concentric with the bearing**

Concept: Petroff assumed a full oil film of uniform thickness equal to the radial clearance c, i.e. zero eccentricity.

With a linear velocity profile the viscous shear gives $f = 2\pi^2(\mu N/p)(r/c)$.

This holds only for light loads at high speed, where the journal runs nearly concentric: option **(C)**.

Exam tip: At heavier loads the journal moves off centre and the full hydrodynamic (Sommerfeld/Raimondi–Boyd) analysis is needed.

Q11 **Answer: 1.77 (accept 1.75 to 1.78)**

Concept: Water power = ρgQH; shaft power = overall efficiency × water power.

Water power = $1000 \times 9.81 \times 2 \times 100 = 1.962 \times 10^6$ W.

Shaft power = $0.9 \times 1.962 =$ **1.77 MW**.

Exam tip: 1 m^3/s falling through 1 m gives about 9.81 kW — a quick check on the order of magnitude.

Q12 **Answer: (B) 60 °C**

Concept: In series the same heat flux passes through each layer, so each temperature drop is proportional to its thermal resistance L/k.

$R_2 = L/k_2$ and $R_1 = L/(2k_2) = R_2/2$, so $\Delta T_2 : \Delta T_1 = 2 : 1$.

$\Delta T_2 = 90 \times 2/3 =$ **60 °C**, option (B).

Exam tip: The poorer conductor always takes the larger share of the temperature drop — the idea behind insulation.

Q13 **Answer: 5**

Concept: For a reversible refrigerator, COP = $T_L/(T_H - T_L)$, with absolute temperatures.

T_L = 250 K, T_H = 300 K.

COP = 250/(300 − 250) = **5**.

Exam tip: Never use °C in this formula: 23/50 or −23/50 are the usual wrong answers.

Q14 **Answer: (D) Gas tungsten arc welding (GTAW/TIG)**

Concept: In a consumable-electrode process the electrode melts and becomes the filler metal.

SMAW (coated stick), SAW (bare wire under flux) and MIG (continuously fed wire) all melt their electrodes.

TIG uses a tungsten electrode that does not melt; filler, if needed, is added separately: option **(D)**.

Exam tip: Plasma arc welding also uses a non-consumable tungsten electrode.

Q15 **Answer: (A) Kanban (just-in-time)**

Concept: In a pull system, a downstream stage's actual consumption triggers production upstream; in a push system, production is released to a plan or forecast.

MRP, forecast-based scheduling and calendar batch release all push work into the shop.

Kanban cards authorise production only when parts are used: **Kanban (JIT)**, option (A).

Exam tip: The number of kanbans N = $D(T_w + T_p)(1 + \alpha)/C$ caps the work-in-process.

What next?

Day 83 test tonight, 7 to 11 PM

65 questions · 180 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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Next, Day 84/90: Formula Sprint: Manufacturing & IE. Comment **GATE90** on our reel to get all links. Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.